

# Products and systems for the protection and repair of concrete structures — Definitions, requirements, quality control and evaluation of conformity —

## Part 1: Definitions

The European Standard EN 1504-1:2005 has the status of a  
British Standard

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# National foreword

This British Standard is the official English language version of EN 1504-1:2005. It supersedes BS EN 1504-1:1998 which is withdrawn.

The UK participation in its preparation was entrusted to Technical Committee B/517/8, Protection and repair of concrete structures, which has the responsibility to:

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## Summary of pages

This document comprises a front cover, an inside front cover, the EN title page, pages 2 to 8, an inside back cover and a back cover.

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English Version

**Products and systems for the protection and repair of concrete  
structures - Definitions, requirements, quality control and  
evaluation of conformity - Part 1: Definitions**

Produits et systèmes pour la protection et la réparation des  
structures en béton - Définitions, prescriptions, maîtrise de  
la qualité et évaluation de la conformité - Partie 1:  
Définitions

Produkte und Systeme für den Schutz und die  
Instandsetzung von Betontragwerken - Definitionen,  
Anforderungen, Güteüberwachung und Beurteilung der  
Konformität - Teil 1: Definitionen

This European Standard was approved by CEN on 2 June 2005.

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## Foreword

This European Standard (EN 1504-1:2005) has been prepared by Technical Committee CEN/TC 104 "Concrete and related products", the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by January 2006, and conflicting national standards shall be withdrawn at the latest by December 2008.

This document supersedes EN 1504-1:1998.

It has been developed by sub-committee 8 "Products and systems for the protection and repair of concrete structures" (Secretariat AFNOR).

This document is part of a series of standards dealing with the general considerations related to the products and systems for the repair and protection of concrete structures. The other parts of this Standard are listed below:

- EN 1504-2, *Products and systems for the protection and repair of concrete structures - Definitions, requirements, quality control and evaluation of conformity - Part 2: Surface protection systems for concrete*
- EN 1504-3, *Products and systems for the protection and repair of concrete structures - Definitions, requirements, quality control and evaluation of conformity - Part 3: Structural and non-structural repair*
- EN 1504-4, *Products and systems for the protection and repair of concrete structures - Definitions, requirements, quality control and evaluation of conformity - Part 4: Structural bonding*
- EN 1504-5, *Products and systems for the protection and repair of concrete structures - Definitions, requirements, quality control and evaluation of conformity - Part 5: Concrete injection*
- prEN 1504-6, *Products and systems for the protection and repair of concrete structures - Definitions, requirements, quality control and evaluation of conformity - Part 6: Anchoring of reinforcing steel bar*
- prEN 1504-7, *Products and systems for the protection and repair of concrete structures - Definitions, requirements, quality control and evaluation of conformity - Part 7: Reinforcement Corrosion Protection*
- EN 1504-8, *Products and systems for the protection and repair of concrete structures - Definitions, requirements, quality control and evaluation of conformity - Part 8: Quality control and evaluation of conformity*
- ENV 1504-9, *Products and systems for the protection and repair of concrete structures - Definitions, requirements, quality control and evaluation of conformity - Part 9: General principles for the use of products and systems*
- EN 1504-10, *Products and systems for the protection and repair of concrete structures - Definitions - Requirements - Quality control and evaluation of conformity - Part 10: Site application of products and systems, and quality control of the works*

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## 1 Scope

This European Standard defines terms relating to products and systems for repair, for use in maintenance and protection, restoration and strengthening of concrete structures.

## 2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

Not applicable.

## 3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

### 3.1 General

#### 3.1.1

##### **batch**

quantity of material made in a single operation, or in the case of continuous production for a defined quantity (in tonnes) which shall be demonstrated by the producer to have a uniform composition and shall not exceed one day's production

#### 3.1.2

##### **declared value**

value declared and documented by the manufacturer for identification or performance requirements

#### 3.1.3

##### **identification test**

test carried out to verify a declared value of the composition or property of the product or system in terms of consistency of the production

NOTE This is to ensure that the product or system being tested corresponds to the product or system subjected to the initial type test, within the permitted tolerances.

#### 3.1.4

##### **performance**

ability of a product or system to provide an effective and durable repair or protection without adverse effects on the original structure, other structures, site operatives, users, third parties and environment

#### 3.1.5

##### **performance requirements**

required mechanical, physical and chemical properties of products and systems to provide durability and stability of both the repaired concrete and the structure

#### 3.1.6

##### **performance test**

test carried out to verify a value for a required property of the product or system in terms of its specified performance during application and use

NOTE This is to ensure that the product or system conforms to its specified performance characteristics.

**3.1.7****product**

constituents formulated for the repair or protection of concrete structures

**3.1.8****systems**

two or more products which are used together, or consecutively, to undertake repair or protection of concrete structures

**3.1.9****technology**

application of a product or system using specific equipment or method (for example crack injection)

**3.2 Main categories of products and systems****3.2.1****anchoring products and systems**

products and systems which:

- anchor reinforcement into concrete to give adequate structural behaviour;
- fill cavities in order to ensure a continuity between steel and concrete elements.

**3.2.2****injection products and systems**

products and systems which, when injected into a concrete structure, restore the structural integrity and/or durability

**3.2.3****non-structural repair products and systems**

products and systems which, when applied on a concrete surface, restore the geometric or aesthetic aspect of the structure

**3.2.4****reinforcement protection products and systems**

products and systems applied to unprotected reinforcement to provide corrosion protection

**3.2.5****structural bonding products and systems**

products and systems applied to concrete to provide a durable structural bond to additional applied material

**3.2.6****structural repair products and systems**

products and systems applied to a concrete structure, to replace defective concrete and to restore structural integrity and durability

**3.2.7****surface protection products and systems**

products and systems which, when applied, improve the durability of concrete and reinforced concrete structures

**3.3 Main chemical types and constituents of protection and repair products and systems****3.3.1****additions**

finely divided inorganic materials that may be added to repair products in order to improve certain properties or to achieve special properties

There are two types of additions:

- nearly inert additions (type I); and
- pozzolanic or latent hydraulic additions (type II).

### 3.3.2

#### **additives for hydraulic binders**

products which are added with hydraulic binder to give specific properties and which are not covered by admixtures and additions

### 3.3.3

#### **additives for reactive polymer**

products other than admixtures and additions which give the repair product specific properties

NOTE Typical additives are, for example:

- plasticizers;
- flexibilizers;
- accelerators;
- retarders;
- materials which regulate the rheology;
- pigments;
- fillers.

### 3.3.4

#### **admixtures**

material added during the mixing process of concrete in a quantity not more than 5 % by mass of the cement content of the concrete, to modify the properties of the mix in the fresh and/ or hardened state

### 3.3.5

#### **coating**

treatment to produce a continuous protective layer on the surface of concrete

NOTE 1 Thickness is typically of 0,1 mm to 5,0 mm. Particular applications may require a thickness greater than 5 mm.

NOTE 2 Binders may be, for example, organic polymers, organic polymers with cement as a filler or hydraulic cement modified with polymer dispersion.

### 3.3.6

#### **hydraulic binders (H)**

inorganic material which reacts with water, undergoing a hydration reaction to produce a solid material

NOTE They are generally cements conforming to EN 197-1 or to EN 413-1, building limes conforming to EN 459-1 or in combination with other cements.

### 3.3.7

#### **hydraulic mortars and hydraulic concretes (CC)**

mortars and concretes based on a hydraulic binder which is blended together with graded aggregates and may include admixtures and additions and which, when mixed with water, set by a hydration reaction

**3.3.8****hydrophobic impregnation**

treatment of concrete to produce a water-repellent surface. The pores and capillaries are internally coated, but they are not filled. There is no film on the surface of the concrete and there is little or no change in its appearance

NOTE Active compounds may be, for example, silanes or siloxanes.

**3.3.9****impregnation**

treatment of concrete to reduce the surface porosity and to strengthen the surface. The pores and capillaries are partially or totally filled

NOTE 1 This treatment usually leads to a discontinuous, thin film on the concrete surface.

NOTE 2 Binders may be, for example, organic polymers.

**3.3.10****polymer hydraulic cement mortars or concretes (PCC)**

hydraulic mortars or concretes modified by the addition of polymer additives, which are added in sufficient quantity to give specific properties

NOTE Polymers typically used include:

- acrylic, methacrylate or modified acrylic resins as redispersible powders or aqueous dispersions;
- vinyl mono-, co- and ter-polymers as redispersible powders or aqueous dispersions;
- styrene butadiene co-polymer, generally as aqueous dispersions;
- natural rubber latices;
- epoxies.

**3.3.11****polymer mortars and polymer concretes (PC)**

blended mixtures of polymer binder and graded aggregates which set by polymerisation reaction

**3.3.12****reactive polymer (P) binder**

binders which generally consist of two components, a reactive polymer base and a hardener or catalyst, and which cure at ambient temperature. Additives (see 3.3.3) may also be added

NOTE 1 Ambient moisture vapour may act as the hardener/catalyst in some systems.

NOTE 2 Typical binders are, for example:

- epoxies;
- unsaturated polyesters;
- cross-linkable acrylics;
- one or two-component polyurethanes.

## Bibliography

- [1] EN 197-1, *Cement – Part 1: Composition, specifications and conformity criteria for common cement*
- [2] EN 413-1, *Masonry cement - Part 1: Composition, specifications and conformity criteria*
- [3] EN 459-1, *Building lime - Part 1: Definitions, specifications and conformity criteria*



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